



UNIVERSITY OF TRIESTE

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Bachelor in Artificial Intelligence and Data Analytics

GRADUATION THESIS

Machine Learning for Predicting the Southward IMF Component

B_z :

A Study Based on Reiss et al. (2021) with Post-2021 Improvements

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Abstract

Predicting the south–north component of the interplanetary magnetic field (IMF) B_z in advance of an Interplanetary Coronal Mass Ejections (ICMEs) arrival, remains a major challenge for space-weather services. This thesis has two main goals. The first one is to summarize through a review the results and the machine learning techniques used by the study *"Machine Learning for Predicting the B_z Magnetic Field Component From Upstream in Situ Observations of Solar Coronal Mass Ejections"*, which uses upstream in-situ data from 348 ICMEs to forecast the minimum B_z and the peak total B_t inside the magnetic obstacle.

The second part explores whether newer deep-learning techniques released after 2021 can push those results further, possibly without slowing real-time use.

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Chapter 1

Introduction

1.1 Physical Background and Geomagnetic Context

1.1.1 Space Weather

Space weather includes changes in the solar wind that can disturb Earth’s magnetic environment and cause problems for satellites, power grids, or communication systems. The solar wind is a stream of charged particles that carries a magnetic field called the **Interplanetary Magnetic Field (IMF)**. When solar wind conditions change abruptly, most notably during a **Coronal Mass Ejection (CME)**, Earth’s magnetosphere can be driven far from equilibrium, producing geomagnetic storms.

1.1.2 Coronal Mass Ejections (CMEs) and Their Structure

A CME is a large eruption of plasma and magnetic field that comes from the Sun. It usually has two parts (Figure 1.1):

1. **Sheath**: a turbulent region of compressed solar wind that comes just after the shock front. This forms only if the CME is fast enough to drive a shock.
2. **Magnetic obstacle** (also called a *flux rope*): a more stable structure with a rotating magnetic field that follows the sheath. Some CMEs, especially slow or poorly structured ones, may lack a clear magnetic obstacle.

The spacecraft is positioned at the **Lagrange Point 1 (L1)**, a gravitationally stable location between the Sun and Earth that allows continuous monitoring of solar wind conditions. As a Coronal Mass Ejection (CME) approaches, the spacecraft first encounters the turbulent *sheath* region, followed by the more organized *magnetic obstacle*. Since the sheath arrives first, it can provide an early indication of the properties of the magnetic structure that follows.¹

1.1.3 Why the IMF Component B_z Matters Most

The IMF component B_z is the north–south part of the magnetic field in GSM coordinates. If B_z is positive (northward), it is aligned with Earth’s magnetic field and does not interact much. If B_z is negative (southward), it can connect with Earth’s field and transfer energy

¹Unless stated otherwise, all IMF values are expressed in **Geocentric Solar Magnetospheric (GSM)** coordinates.

into the magnetosphere [2]. This process is called **magnetic reconnection** and can build up energy that later causes geomagnetic storms [3]. In practice, the minimum B_z and the maximum transverse magnetic field B_t are used as simple indicators of how strong the interaction could be.

1.1.4 Geomagnetic Storms

If B_z stays southward for a long time, the magneto-tail stretches and stores energy. Later, it releases this energy by reconnecting magnetic field lines in the tail. This sends high-energy plasma toward Earth and causes strong electric currents in the upper atmosphere. This process creates geomagnetic storms, which can cause auroras, satellite damage, and power grid failures.

1.1.5 Quantifying Disturbance: The Kp Index

The Kp index is a number from 0 to 9 used to describe how disturbed Earth's magnetic field is. It is based on measurements from magnetometers at mid-latitudes and updates every three hours. High Kp values are linked to auroras, GPS problems, satellite drag, and transformer overheating. Since Kp reacts quickly to sustained southward B_z , predicting B_z in advance is one of the best ways to warn about upcoming geomagnetic activity.

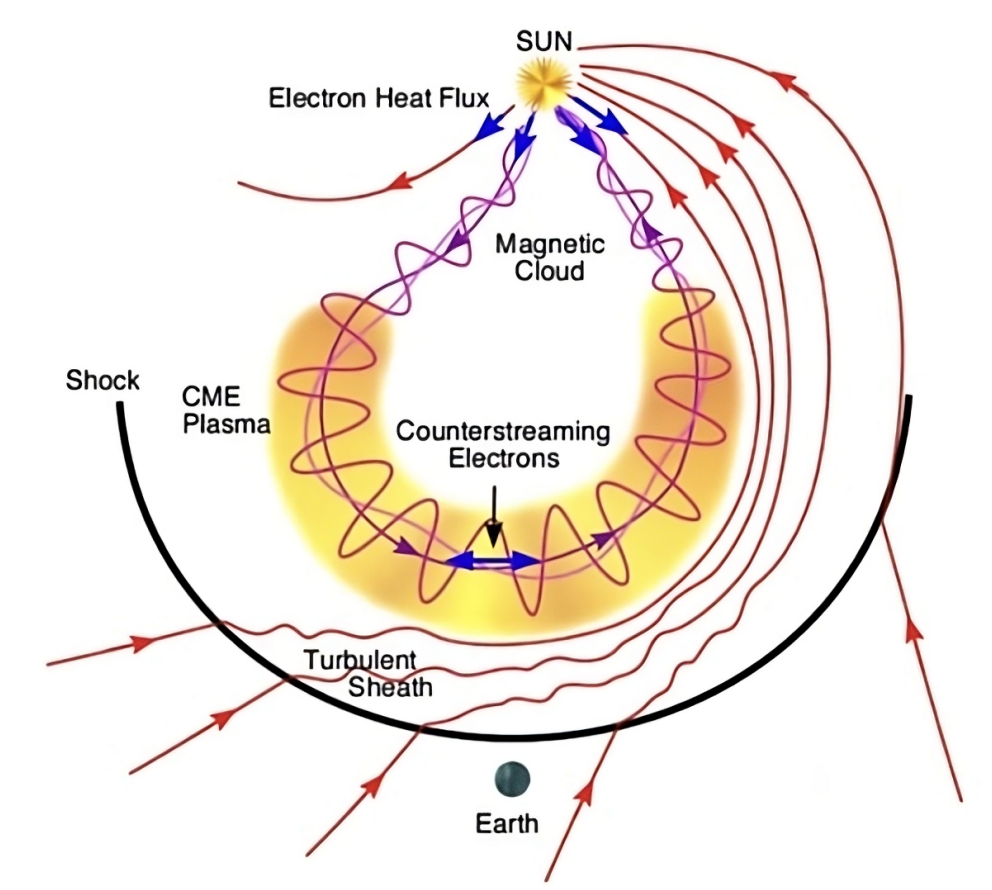


Figure 1.1: Illustration of a CME propagating from the Sun to Earth. The structure includes a leading shock, a turbulent sheath region, and a magnetic cloud characterized by counter-streaming electrons and electron heat flux. Adapted from [4].

1.2 The B_z Forecasting Problem

The interplanetary magnetic-field component B_z , expressed in **Geocentric Solar Magnetospheric (GSM)** coordinates, is defined as positive when pointing northward and negative when pointing southward. A sustained southward B_z orientation enables magnetic reconnection at the day-side magnetopause, facilitating the transfer of solar-wind energy into Earth’s magnetosphere. This energy transfer drives geomagnetic storms, which are tracked by indices like D_{st} and K_p [3]. Even a few nanotesla of negative B_z can trigger auroras, radio blackouts, or transformer failures.

However, predicting B_z in advance is difficult, often referred to as the B_z *problem*, because the IMF structure cannot be directly measured until the solar wind reaches near-Earth space. Monitors at the Sun–Earth **Lagrange Point 1 (L1)** (e.g., DSCOVR) offer only 30–60 minutes of lead time, and their single-point sampling misses key spatial features, such as CME flux-rope rotations. Physics-based **Magnetohydrodynamic (MHD)** models are self-consistent but require detailed inputs and struggle to resolve the turbulent sheath region ahead of CMEs, often resulting in large errors.

To address these limitations, researchers have turned to **Machine Learning (ML)**. Unlike MHD models, ML methods do not require explicit physical equations. Instead, they learn statistical patterns from past data, making them especially useful when inputs are sparse or uncertain. Reiss et al. [11] demonstrated that a **gradient-boosted tree model (XG-Boost)**, trained on OMNI solar-wind measurements, can forecast the minimum B_z inside a CME with a **Mean Absolute Error (MAE)** of 3.12 nT and a **Pearson Correlation Coefficient (PCC)** of 0.71, achieving a lead time ² of up to 60 minutes.

The data source that will be used in this thesis is the same as [11]: 1-minute solar-wind and IMF data from the **Wind spacecraft**, processed via NASA’s **OMNIWeb** system and time-shifted to Earth’s bow shock. Performance is evaluated using MAE (error magnitude) and PCC (trend agreement).

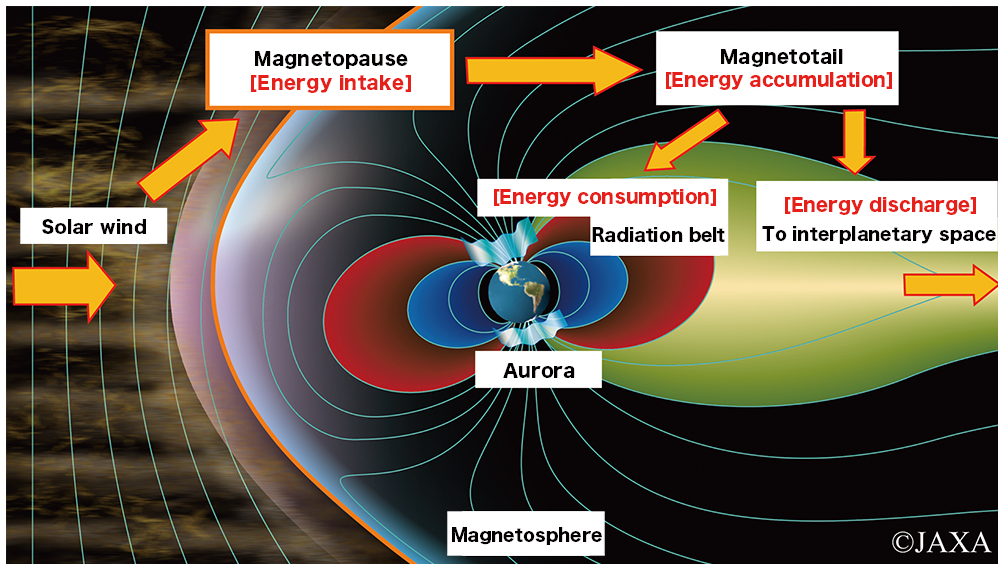


Figure 1.2: *Solar-wind–magnetosphere coupling during sustained southward B_z . Day-side reconnection channels solar-wind energy into the magnetosphere. Adapted from [5].*

²Lead time refers to the interval between when the sheath is observed at L1 and when the minimum B_z occurs within the magnetic obstacle.

Chapter 2

Machine Learning Techniques used

In their study, Reiss et al. (2021) compared three different supervised regression models to predict key magnetic field properties inside ICMEs. A simple **Linear Regressor (LR)** was used as a baseline for benchmarking. A more advanced **Random Forest Regressor (RFR)** was evaluated as a robust ensemble method. However, the best performing approach was the **Gradient Boosted Regressor (GBR)**, implemented via *XGBoost*, which was ultimately selected for the final predictive tool due to its superior accuracy and interpretability.

2.1 Overview of Supervised Regression

Supervised regression is a type of machine learning in which the goal is to learn a function that maps input variables X to a continuous output variable y , based on example pairs (X_i, y_i) . The algorithm is trained on a labeled dataset, where each input has an associated target value. Once trained, the model can be used to predict outcomes for new, unseen data.

In the context of space weather forecasting, this means mapping solar wind and magnetic field observations to quantitative predictions such as the minimum of B_z or the peak of B_t within an ICME. The models compared in Reiss et al. (2021) fall into this category, as they were trained to predict continuous values from structured input features extracted from in situ measurements.

2.2 Linear Regression

Linear regression is one of the simplest and most interpretable forms of supervised learning. It assumes a linear relationship between the input features and the target variable, and attempts to fit a model of the form:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

where $\hat{y}$ is the predicted value, x_i are the input features, and β_i are the model coefficients.

In Reiss et al. (2021), linear regression was included primarily as a baseline model. While easy to implement and fast to compute, its performance was clearly limited due to its inability to capture non-linear interactions between variables. On the test set, it produced a higher

mean absolute error and lower correlation compared to the tree-based methods. Nonetheless, it serves as a useful benchmark to evaluate whether more complex models provide meaningful improvements.

2.3 Random Forest

Random Forest is an ensemble learning method based on decision trees. The idea is to combine the predictions of many individual trees, each trained on a random subset of the data and a random subset of features, to obtain a more robust and accurate model. The final output is given by averaging the predictions from all trees (in regression tasks) or by majority vote (in classification).

For regression, the output of the Random Forest is given by:

$$\hat{y}_{\text{RF}}(x) = \frac{1}{M} \sum_{m=1}^M T_m(x)$$

where $T_m(x)$ is the prediction of the m -th tree, and M is the total number of trees in the ensemble.

This randomness helps reduce overfitting, which is a common problem in single decision trees. Each tree may capture different patterns or noise, but their ensemble average tends to generalize better to unseen data.

In the Reiss et al. (2021) study, the Random Forest Regressor (RFR) was used as a strong non-linear baseline. Compared to linear regression, it was much more effective at modeling complex interactions between solar wind variables. In terms of performance, it achieved a significantly lower mean absolute error and higher correlation coefficient with the observed B_z values.

The RFR's strength lies in its ability to provide high quality predictions with relatively little tuning, and its interpretability through measures such as feature importance.

2.4 Gradient Boosted Trees (XGBoost)

Gradient Boosted Trees are an ensemble method where multiple decision trees are trained sequentially, with each new tree correcting the errors of the previous ones. This process, known as boosting, allows the model to improve performance iteratively by focusing on the most difficult examples. XGBoost is a highly optimized implementation of this technique that supports regularization, missing value handling, and parallelized computation.

The prediction of a Gradient Boosted model is a weighted sum of the outputs of all the trees:

$$\hat{y}(x) = \sum_{m=1}^M \gamma_m T_m(x)$$

where $T_m(x)$ is the output of the m -th tree, γ_m is its learning rate (or weight), and M is the total number of trees.

In the context of the article of Reiss et al.(2021), XGBoost proved to be the most effective model among those tested. Its ability to handle structured tabular input, model complex non-linear relationships, and produce fast predictions made it a suitable candidate for space weather forecasting tasks. The detailed reasons behind this choice are discussed more formally in **Section 3.4.1**.

XGBoost achieved the best validation performance, both in terms of mean absolute error and correlation with the observed values. It also showed strong classification skill when identifying extreme geoeffective events. These qualities justified its selection as the core model of the final predictive pipeline.

Chapter 3

State of the art

3.1 Legacy Forecasting Approaches Before 2021

Forecasting the south-north IMF component B_z has a long history that predates modern ML solutions. Early approaches to B_z forecasting included: empirical formulas, physics-based MHD simulations, and initial data-driven techniques, such as regression or statistical fits. Understanding their strengths and limitations gives context to the tree-based model later adopted by Reiss et al. [11].

3.1.1 Empirical and Persistence Models

The simplest operational tools assume that the solar wind conditions measured at L1 will remain constant over time, a method known as *persistence*, or apply linear regressions between upstream quantities and geomagnetic indices such as D_{st} or K_p . These heuristics are computationally inexpensive, but their predictive skill drops sharply once the flux rope’s interior begins to rotate away from the magnetic field observed in the sheath. Moreover, single-point measurements cannot capture the three-dimensional changes in ICME orientation during propagation, often leading to sign errors in B_z .

3.1.2 Physics-Based MHD Simulations

Full-physics solvers such as: SUSANOO-CME, AWSOM-SWMF, MAS and EUHFORIA, evolve a CME from the low corona to 1 AU, preserving magnetic tension forces and flux-rope topology [15, 6, 16, 10]. In principle, these codes can reproduce the entire $B_z(t)$ profile; in practice, they face two obstacles:

- a) **Uncertain initial conditions.** Even small errors in the input data at the edge of the heliosphere can grow into large differences by the time the ICME reaches Earth, as shown by ensemble simulations. [Kay2018, Moestl2018].
- b) **Computational cost.** Even reduced-resolution runs may take hours, which is too slow for <1 h warning during severe storms [11].

Consequently, operational centres still rely on simpler proxies for B_z , while MHD runs serve mainly as research or situational-awareness tools. The persistent mismatch between desired lead time and achievable fidelity is a key aspect of the B_z *problem*.

3.1.3 Rule-Based Flux-Rope Classification

A parallel line of research used morphological clues to guess the flux-rope field orientation. Early flux-rope models proposed by **Chen1996**, **Chen1997** exploited the coherence of large magnetic clouds to forecast their internal structure up to ~ 10 h ahead. Later, category schemes combined coronagraph imagery with empirical lookup tables [1, 14], while solar-wind monitoring, refined the prediction along the way [7]. Although being empirically useful, these methods often fail for distorted or weak magnetic structures, especially during solar maximum, when such ICMEs are more common.

3.1.4 First-Generation Data-Driven Techniques (2014–2020)

Around 2015, the field began experimenting with automated pattern recognition and analogue matching:

- **Pattern recognition:** decision rules learned from historical events to assign a likely B_z class [12].
- **Past analogues:** searching archived solar-wind segments with similar plasma and field signatures to the present case [9].
- **Deep learning prototypes:** convolutional networks trained to detect CME related patterns in fast-sampled solar wind data [13].

These studies reduced human subjectivity and handled noisy inputs better than hand-crafted rules, but they shared two disadvantages:

- a) Limited event counts (< 400 ICMEs) constrained the depth of usable neural networks.
- b) Most models treated the entire magnetic obstacle as a single target, ignoring how its properties change over time and not accounting for uncertainty in the predictions.

Summary and Open Issues

- **Empirical and persistence models** are fast but lose skill beyond ~ 60 min lead time.
- **MHD simulations** preserve physics, yet depend on uncertain boundary conditions and long runtimes.
- **Rule-based and early ML approaches** add quantitative structure but, remain data-limited and often ignore time ordering.

These gaps, especially the need for rapid, interpretable forecasts from sparse observations, set the stage for the gradient-boosted tree strategy proposed by Reiss et al. [11].

3.2 Dataset and Event Selection in Reiss et al. [11]

Accurate B_z forecasts require a representative set of ICME observations near 1 AU, where in-situ measurements reflect the true variety of magnetic structures. Reiss et al. [11] used the open *ICMECAT v2.0* catalog (last updated on 29 April 2021), which lists 558 ICMEs observed by the Wind and STEREO spacecraft between 2007 and 2021.

3.2.1 Spacecraft and Instruments

- **Wind:** Magnetic Field Investigation (MFI) and Solar Wind Experiment (SWE) plasma instruments.
- **STEREO-A/B:** IMPACT magnetometers and PLASTIC plasma analyzers.

All time series are resampled to a common 1-minute cadence using NASA’s OMNI processing pipeline, then time-shifted to the Earth’s bow shock, so that sheath arrival times are consistent across spacecraft.

3.2.2 Quality Filters

From the 558 candidates, Reiss et al. applied three successive criteria:

- Presence of a Magnetic Obstacle (MO) with coherent rotation, as defined by Nieves-Chinchilla et al. [8].
- Detection of either a distinct sheath or a clear density pile-up ahead of the MO.
- Fewer than 5% missing samples in the magnetic-field or plasma variables used for feature engineering.

After filtering, the study retains **348 ICMEs**: 149 from Wind, 123 from STEREO-A, and 76 from STEREO-B.

3.2.3 Statistical Snapshot of the Final Sample

The table below shows the average duration and characteristic field values across the 348 ICMEs used in the study.

Table 3.1: *Mean $\pm$ standard deviation of ICME properties in the sheath and magnetic obstacle.*

Property	Sheath	Magnetic Obstacle
Duration [h]	9.0 ± 6.0	25.2 ± 14.6
Peak B_t [nT]	-	14.3 ± 7.6
Minimum B_z [nT]	-	-9.4 ± 6.8

Figure 1 of the original paper shows the full probability distributions of these quantities and confirms that extreme events (i.e., $|B_z| > 20$ nT) do occur, but are relatively rare.

3.2.4 Segmentation: Sheath vs. Magnetic Obstacle

Event times are taken directly from ICMECAT fields:

$$\text{Sheath: } t_{\text{ICME_start}} \rightarrow t_{\text{MO_start}} \mid \text{MO: } t_{\text{MO_start}} \rightarrow t_{\text{MO_end}}.$$

This clear partition supports two forecast horizons:

- *Pure-sheath* mode (lead time ≈ 10 h).
- *Progressive* mode adding the first n hours of the MO ($n = 1 \dots 4$) to test how early partial MO data improves skill.

3.2.5 Train and Test set

The cleaned dataset is randomly split 70% / 30% (243 / 105 events) into training and blind-test sets, with five-fold cross-validation performed within the training subset. No temporal ordering is imposed, as each ICME is treated as an independent sample; a defensible choice given the multi-year timespan of the data.

3.3 Feature Engineering

Following Reiss et al. [11], each ICME is converted from raw 1-minute time series into a compact vector of **42 scalar features** (7 physical variables $\times$ 6 summary statistics). The goal is to retain the key patterns a human forecaster would typically examine, such as trends, variability, and extrema, while keeping the input compact enough to allow inference within milliseconds.

3.3.1 Base Variables

- B_t : total magnetic-field strength
- B_x, B_y, B_z : GSM components
- v_p : proton flow speed
- N_p : proton number density
- T_p : proton temperature

These seven variables capture both field geometry (B) and plasma dynamics (v_p, N_p, T_p) that influence shock strength and magnetic compression ahead of the flux rope.

3.3.2 Summary Statistics per Variable

- 1) **Mean** (μ): represents the average value over the window, indicating the general background state.
- 2) **Standard deviation** (σ): quantifies short-term fluctuations around the mean, often associated with turbulence or wave-like activity.
- 3) **Minimum**: captures the most negative value within the window, e.g., the deepest southward dip in B_z , or the lowest density or field strength.
- 4) **Maximum**: identifies the highest observed value, such as peak magnetic compression or the strongest northward deviation.
- 5) **$\frac{\text{Max}}{\text{Min}}$ ratio**: expresses the contrast between extremes; useful for detecting sudden, sharp transitions in the data.
- 6) **$\frac{\text{Mean}}{\sigma}$ ratio**: serves as a signal-to-noise indicator; higher values suggest a smooth, consistent trend, while lower values reflect noisy or irregular structures.

3.3.3 Reason Behind these Six Metrics

- They are *fast* to compute in streaming mode.
- Each one of them maps to an interpretable physical aspect: peaks relate to shock compression, the mean to background polarity, variability to sheath turbulence.
- Together they proved sufficient to reach the lowest validation error in an ablation test reported by Reiss et al. [11].
- Adding higher-order moments (skewness, kurtosis) showed no significant gain given only 348 events.

So, the feature matrix is built in this way: 7 variables $\times$ 6 statistics = 42 inputs. This 42-dimensional feature vector forms the input to the gradient-boosted-tree model discussed in **Section 3.4**.

3.4 Model Choice and Training Protocol

3.4.1 Why Gradient-Boosted Trees?

Reiss et al. evaluated several regressors and found that **XGBoost**, an efficient implementation of Gradient-Boosted Decision Trees (GBDT), performed the best validation skill for three main reasons:

- a) Performs reliably on *tabular* datasets with variables on different scales and limited sample sizes. Conditions where deep neural networks are more prone to overfit.
- b) Automatically captures non-linear interactions between features (e.g., how variability in v_p combines with peaks in B_t) without the need for manual feature engineering.
- c) Offers built-in estimates of feature importance, which help interpret model behavior and identify the most influential physical variables.

3.4.2 Hyper-parameter Grid

The model was tuned using grid search across the following parameter ranges. The final selected values after cross-validation are shown in the rightmost column.

Table 3.2: *XGBoost hyper-parameters: search space and selected values.*

Parameter	Search Range	Selected Value
n_estimators	100 - 600	350
max_depth	3 - 8	6
learning_rate	0.01 - 0.2	0.05
subsample	0.6 - 1.0	0.8
colsample_bytree	0.5 - 1.0	0.8
reg_lambda	0 - 5	1.0

The optimal configuration converged at `n_estimators = 350`, `max_depth=6`, `learning_rate = 0.05`, with 80% row and 80% column subsampling, balancing bias and variance on the 5-fold cross-validation surface.

3.4.3 Cross-Validation Design

- **Split:** 70% training (243 events), 30% blind test (105 events).
- **Inner loop:** stratified 5-fold CV within the training pool for hyper-parameter tuning.
- **Early stopping:** training halts if validation MAE does not improve for 30 rounds, preventing over-fitting.

3.4.4 Inference Speed

Running on a single CPU core, the trained model predicts one ICME window in ≈ 2 ms, well below the 60s data cadence, making it suitable for operational space-weather pipelines.

With the model calibrated, we now assess its quantitative performance across different lead times (**Section 3.5**) and highlight the remaining limitations that motivate the use of more recent post-2021 techniques.

3.5 Performance Across Lead Times

3.5.1 Forecast Targets and Metrics

The model predicts two quantities derived from the magnetic obstacle:

- **Minimum B_z :** most geoeffective southward excursion
- **Peak B_t :** maximum total field, linked to compression

Skill is assessed using two complementary approaches:

- Continuous metrics:** Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Skill Score (SS), and Pearson Correlation Coefficient (PCC).
- Event-based verification:** an ICME is considered an event if it exceeds a data-driven threshold, $\min(B_z) < -9.06$ nT or $\max(B_t) > 13.89$ nT, corresponding to the mean values of the test set. From this, a contingency table yields: Threat Score (TS), True Positive Rate (TPR), False Positive Rate (FPR), True Skill Statistic (TSS), and Bias Score (BS).

3.5.2 Continuous Metrics (Sheath + 4h of the MO, 105 ICMEs)

Table 3.3: Continuous validation metrics for three learners on the test set. Values reproduced from Table 3 of [11].

Target	Model	MAE	RMSE	SS	PCC
$\min(B_z)$	LR	4.80	11.97	-2.24	0.32
	RFR	3.16	4.73	0.49	0.70
	GBR	3.12	4.77	0.49	0.71
$\max(B_t)$	LR	3.64	9.48	-0.55	0.54
	RFR	2.41	3.79	0.75	0.88
	GBR	2.23	3.20	0.82	0.91

The Gradient-Boosted Regressor (GBR) yields the lowest MAE of 3.12 nT for $\min(B_z)$ and 2.23 nT for $\max(B_t)$, while maintaining a robust correlation (PCC of 0.71 and 0.91) with the observed values.

3.5.3 Event-Based Skill

Table 3.4: Contingency counts and Binary Skill metrics, reproduced from Table 4 of [11].

Target	Model	N_{obs}	N_{pred}	TP	FP	FN	TN	TPR	FPR	TS	TSS	BS
$\min(B_z)$	LR	60	60	39	21	21	24	0.65	0.47	0.48	0.18	1.00
	RFR	60	67	46	21	14	24	0.77	0.47	0.57	0.30	1.12
	GBR	60	68	46	22	14	23	0.77	0.49	0.56	0.28	1.13
$\max(B_t)$	LR	39	41	28	13	11	53	0.72	0.20	0.54	0.52	1.05
	RFR	39	34	28	6	11	60	0.72	0.09	0.62	0.63	0.87
	GBR	39	37	30	7	9	59	0.77	0.11	0.65	0.66	0.95

With the GBR, the Threat Score reaches 0.56 for strong southward $\min(B_z)$ events and 0.65 for high $\max(B_t)$ events, outperforming both linear regression and the random forest baseline.

3.5.4 Lead-Time Trade-off

Figure 3.1, a direct reproduction of Fig. 3 in Reiss et al. [11], shows two clear tendencies:

- **MAE shrinks as more of the magnetic obstacle is observed:** using only the sheath (0 h) the error is ≈ 3.5 nT, but it drops to $\lesssim 2.4$ nT once 15 h of data have elapsed, so when the forecast lead time is shortest.
- **Correlation strengthens with added context:** the Pearson coefficient for $\min(B_z)$ rises from ≈ 0.60 (sheath-only) to $\gtrsim 0.85$ after 6–8 h, while $\max(B_t)$ reaches a plateau near 0.90 after the first 4 hours.

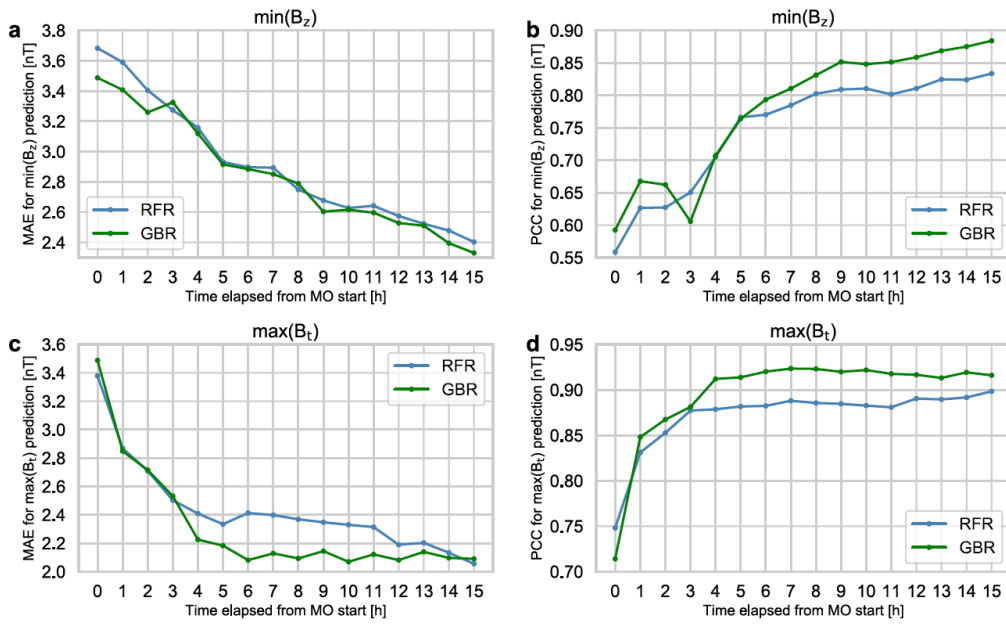


Figure 3.1: Forecast skill versus elapsed time inside the magnetic obstacle (reproduced from Reiss et al. [11]). Panels (a)–(d) show how MAE and PCC for $\min(B_z)$ and $\max(B_t)$ evolve as the model ingests 0 to 15 h of downstream measurements; larger elapsed times correspond to shorter operational lead times.

Thus, while very-long lead-time forecasts (sheath-only) carry ~ 3 – 3.5 nT uncertainty, the model still captures the correct trend and quickly approaches high correlation once partial flux-rope data become available, an essential property for real-time space-weather warnings.

3.5.5 Extreme Events and Limits

For the severe southward cases ($\min(B_z) < -9$ nT) the gradient-boosted tree still attains a Threat Score of ≈ 0.56 , but skill drops further for the very rare, long-duration ICMEs (e.g. the 15 strongest events with $\min(B_z) < -13.9$ nT or $\max(B_t) > 19$ nT). Their scarcity in the data set (only ~ 10 – 15% of the test examples) and the reliance on single-point sampling at L1 probably explain the degraded performance.

Key Takeaways

The gradient-boosted tree (GBR):

- Delivers an MAE of ≈ 3.5 nT for $\min(B_z)$ at the longest operational lead time (~ 10 h, sheath only), declining to < 2.5 nT once a few hours of flux-rope data are available.
- Detects high-risk events with Threat Scores of 0.56 for strong southward $\min(B_z)$ and 0.65 for large $\max(B_t)$, outperforming both linear and random-forest baselines.
- Still struggles with the rarest, long-duration or highly rotating ICMEs (≈ 10 – 15% of the test set), highlighting the need for multi-point observations or sequence-aware models.

3.6 Strengths, Limitations, and Open Gaps

3.6.1 Strengths of the 2021 Pipeline

- **Operational speed:** inference ≈ 2 ms on a single CPU, faster than the 1 min data cadence.
- **Small input size:** only 42 summary features are used, which keeps memory and data transfer requirements low. This makes it easy to run the model on a satellite or at a ground station.
- **Easy to interpret:** tree-based models rank the most important input features. For example, peaks in v_p and B_t are top contributors, matching what is expected from physics.
- **Good performance:** the model achieves under 4 nT error for $\min(B_z)$ and detects strong events with Threat Scores between 0.55 and 0.65, clearly beating simple baselines like persistence or linear regression.

3.6.2 Key Limitations

- **Limited temporal context:** scalar statistics summarize the data but discard time-ordering, so internal rotations and gradual transitions within the flux rope are not captured.
- **Small training set:** with only 348 labeled ICMEs, the model is limited in complexity and may generalize poorly to rare or extreme cases.
- **Single-point view:** the ICME is sampled at just one location (where the spacecraft crosses it at L1), so the model lacks information about its 3D structure. Performance drops when the cloud contains multiple flux ropes or is highly distorted.
- **No uncertainty estimates:** the model outputs a single prediction but gives no confidence level, making it harder for forecasters to assess the reliability of each forecast.

Chapter 4

Implementation of the CNN and GBR

4.1 Convolutional Neural Network (CNN) for Temporal Prediction

In this experiment, we implemented a 1D Convolutional Neural Network (CNN) to predict two critical quantities: the *minimum southward* B_z and the *maximum total magnetic field* B_{tot} within ICMEs. Unlike the models in [11], which used fixed statistical features and did not account for relationships between them, this CNN is able to learn patterns by applying convolutional filters across the input features.

Stratified Preprocessing

To ensure balanced training across different levels of geoeffectiveness, we applied a stratified train/test split based on the distribution of the target variable. Specifically, we divided the events into 5 quantile-based bins using `qcut`, and stratified the sampling accordingly. This procedure avoids training/test imbalances, particularly relevant when predicting $\min(B_z)$, where the distribution is skewed toward weak events (near 0 nT).

The input data included 42 handcrafted features for each event, calculated over a fixed interval (the sheath plus 4 hours). These features were standardized using the `StandardScaler` and included statistical summaries: mean, maximum, minimum, standard deviation, coefficient of variation, and min/max ratio; computed for key solar wind variables: B_x , B_y , B_z , total magnetic field B_{tot} , velocity v_t , density n_p , and temperature t_p .

Before training, the feature matrix was reshaped to the format `(samples, timesteps, 1)` to match the input expected by Conv1D layers. In this structure, *timesteps* refers to the 42 input features for each sample.

4.1.1 Hyper-parameter Grid

The model was tuned using grid search across the following parameter ranges. The final selected values for the 2-layer CNN were chosen based on 3-fold cross-validation with validation MAE as the objective.

Parameter	Search Range	Selected Value
filters	32, 64	64
kernel_size	2, 3, 4	3
dense_units	50, 100, 150	50
batch_size	8, 16	8

Table 4.1: *CNN hyper-parameters: search space and selected values based on 3-fold cross-validation.*

The optimal configuration converged at 64 filters, kernel size 3, and 50 dense units with a batch size of 8. This setup was consistently applied to both the $\min(B_z)$ and $\max(B_{tot})$ regression tasks.

4.1.2 CNN Architecture

The model is composed of two convolutional blocks followed by a dense head. Each block includes a 1D convolution layer followed by max pooling, allowing the model to capture short- and intermediate-scale dependencies across input features; basically it aggregates features and improves generalization.

- **First Conv1D Layer:**
 - 64 filters, kernel size 3, ReLU activation
- **First MaxPooling1D Layer:**
 - Pool size 2
- **Second Conv1D Layer:**
 - 128 filters, kernel size 3, ReLU activation
- **Second MaxPooling1D Layer:**
 - Pool size 2
- **Flatten Layer:**
 - Converts the 3D output into a 1D feature vector (batch size, 128×9)
 - Enables dense layer processing
- **Dense Layer:**
 - 50 units, ReLU activation
 - Combines learned features to estimate the target
- **Output Layer:**
 - 1 unit, linear activation
 - Outputs the predicted value: $\min(B_z)$ or $\max(B_{tot})$

Training Protocol

The model was trained on the stratified training set with the following configuration:

- Optimizer: Adam
- Loss function: Mean Squared Error (MSE)
- Batch size: 8

- Epochs: up to 50
- Early stopping: patience = 5, restore best weights
- Validation split: 10%
- Random seed: 42 (Same as the article)

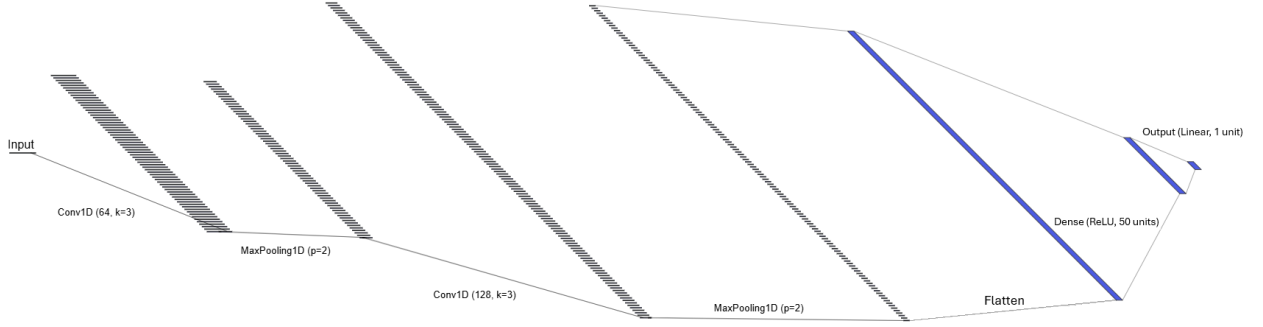


Figure 4.1: Architecture of the 2-layer 1D Convolutional Neural Network used for $\min(B_z)$ and $\max(B_{tot})$ prediction. Each block captures patterns among the 42 statistical features per event.

4.1.3 Evaluation Results

The model was evaluated on a held-out test set (30% of all events), stratified by the quantiles of the target. Results show strong performance for both prediction tasks:

Target	Test MAE (nT)	Test RMSE (nT)	Test PCC	Skill Score
$\min(B_z)$	3.88	5.24	0.76	0.55
$\max(B_{tot})$	1.10	1.98	0.98	0.96

Table 4.2: Performance of the 2-layer CNN on the test set. Results show strong agreement with true values, especially for $\max(B_{tot})$.

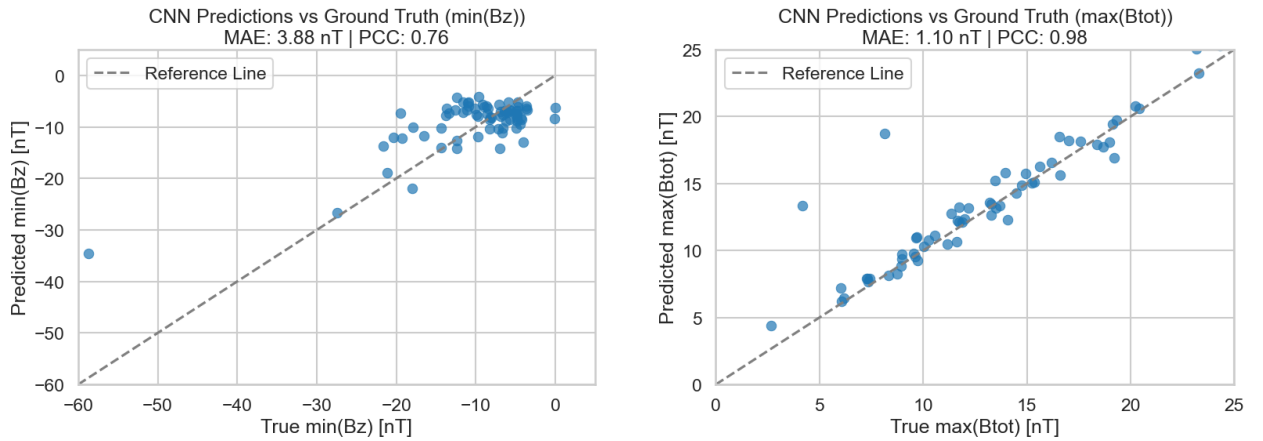


Figure 4.2: Predicted vs true values on the test set for the two regression targets. The model shows good correlation for both, especially for $\max(B_{tot})$. Dashed lines represent the ideal 1:1 correspondence.

These results are competitive with those reported in [11], where the best Gradient Boosting Regressor achieved a PCC of approximately 0.71 for $\min(B_z)$ and 0.91 for $\max(B_{\text{tot}})$. In particular, the CNN performed really well on the prediction of $\max(B_{\text{tot}})$, with an MAE of 1.10 nT and a PCC of 0.98, indicating high consistency between predicted and observed magnetic field magnitudes. This strong performance on $\max(B_{\text{tot}})$ may be attributed to the more regular and positive distribution of the target, which makes it easier for the CNN to model. Moreover, convolutional filters likely helped in capturing dependencies across input variables, and the use of stratified sampling ensured balanced exposure to rare and intense events, enhancing the model’s generalization ability.

4.2 Gradient Boosting Regressor (XGBoost)

In this experiment, we implemented a Gradient Boosting Regressor (GBR) using the XGBoost library to predict two critical quantities within ICMEs: the *minimum southward* B_z and the *maximum total magnetic field* B_{tot} . This model reproduces the feature-based approach introduced in [11], which showed strong predictive performance for B_z forecasting using decision-tree ensembles.

Unlike deep learning models, XGBoost does not learn from raw temporal sequences, but instead operates on handcrafted features. Its strength lies in its ability to model complex non-linear relationships with low computational cost and minimal preprocessing. Gradient boosting combines multiple weak learners (shallow trees) into a powerful ensemble by iteratively correcting the residuals of previous models.

Stratified Preprocessing

The same stratified preprocessing used for the CNN was adopted here. Events were split into five quantile-based bins using `qcut` to ensure balanced representation of both weak and strong events.

Features were standardized using `StandardScaler`, resulting in a fixed-length feature vector of shape `(samples, features)`—built from statistical summaries over the sheath and the first 4 hours of the magnetic obstacle.

4.2.1 Prediction of $\min(B_z)$ and $\max(B_{\text{tot}})$

We trained an XGBoost regressor to predict both the minimum southward component of the IMF ($\min(B_z)$) and the peak total magnetic field strength ($\max(B_{\text{tot}})$) inside the magnetic obstacle. This model follows the same preprocessing used by [11], with 42 statistical features extracted from the sheath and the first 4 hours of the magnetic obstacle, including summary statistics (mean, standard deviation, min, max, etc.) across seven key solar wind and IMF variables.

The model was trained using the following hyperparameters, taken from the best configuration in [11]:

- `n_estimators = 350`
- `max_depth = 6`
- `learning_rate = 0.05`
- `subsample = 0.8`
- `colsample_bytree = 0.8`
- `reg_lambda = 1.0`

While the overall model architecture and feature engineering mirror the original approach by [11], one key improvement introduced in this work is the use of a stratified sampling strategy. By ensuring that both training and test sets contain a balanced distribution of weak and strong events, this adjustment led to a modest but consistent increase in predictive performance, especially for $\min(B_z)$, which is typically dominated by near-zero values. In our case, the Pearson correlation for $\min(B_z)$ improved from 0.71 (as reported by Reiss) to 0.80, indicating a better match between predicted and observed values.

The model was evaluated on a held-out test set (30% of all events), stratified by the quantiles of the target. Results show solid performance for both prediction tasks, with particularly accurate estimates for $\max(B_{\text{tot}})$.

Target	Test MAE (nT)	Test RMSE (nT)	Test PCC	Skill Score
$\min(B_z)$	3.08	4.82	0.80	0.36
$\max(B_{\text{tot}})$	0.79	3.29	0.96	0.88

Table 4.3: Performance of the XGBoost model on the test set. Results indicate strong prediction ability, particularly for $\max(B_{\text{tot}})$, which is less noisy and easier to model.

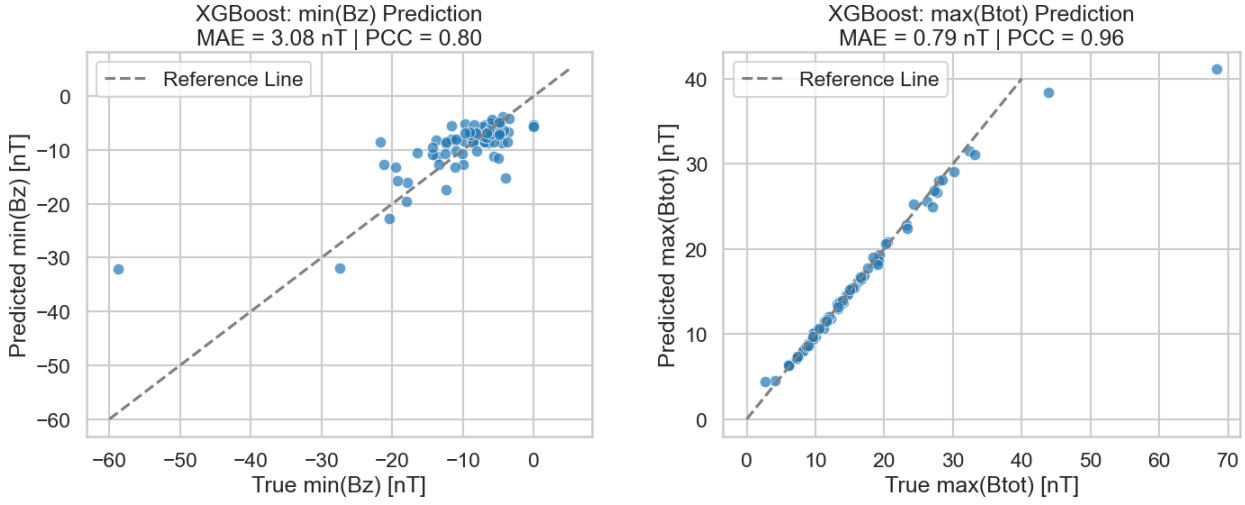


Figure 4.3: Predicted vs true values on the test set for the two regression targets using the XGBoost model. The model shows strong performance on both quantities, with particularly accurate predictions for $\max(B_{\text{tot}})$. Dashed lines represent the ideal 1:1 correspondence.

The results for $\min(B_z)$ are comparable to those reported by [11] (MAE = 3.12 nT, PCC = 0.71), while predictions for $\max(B_{\text{tot}})$ are even more accurate, likely due to its smoother temporal evolution and more symmetric distribution. Scatter plots showing predicted vs true values for both targets confirm the high correlation and general reliability of the model across the test examples.

These strong results validate the suitability of tree-based models for real-time operational settings, where low latency and high interpretability are crucial. In the next section, we will extend this approach using an ensemble of XGBoost models to estimate prediction uncertainty.

4.3 Uncertainty Estimation with Ensemble XGBoost

To enrich the predictive framework with a notion of confidence, we extended the Gradient Boosting Regressor by training an ensemble of XGBoost models. While single models provide point estimates, ensembling allows us to assess variability across predictions and construct empirical confidence intervals. This is particularly useful in operational forecasting, where understanding the uncertainty of a space weather event is just as important as predicting its expected strength.

The ensemble was composed of 20 XGBoost models, each initialized with a different random seed and trained independently on the same stratified dataset. The model architecture and hyperparameters remained fixed across all runs, following the same configuration adopted in the single-model experiment.

At inference time, predictions from all models were collected for each test event. The mean served as the final point estimate, while the 10th and 90th percentiles defined the bounds of the confidence interval. Evaluation was performed on the same 30% stratified test set used previously.

Target	Ensemble MAE (nT)	RMSE (nT)	PCC	Skill Score
$\min(B_z)$	3.07	4.84	0.80	0.36
$\max(B_{\text{tot}})$	0.78	3.34	0.96	0.88

Table 4.4: Performance of the ensemble XGBoost model on the test set. Results are in line with the single-model setup, while offering the additional benefit of confidence intervals.

As expected, the ensemble achieves nearly identical performance to the single XGBoost model. This confirms that most of the variability in predictions stems from the input data rather than model initialization. Nevertheless, the ensemble setup allows us to attach uncertainty estimates to each prediction, providing a richer output for potential operational use.

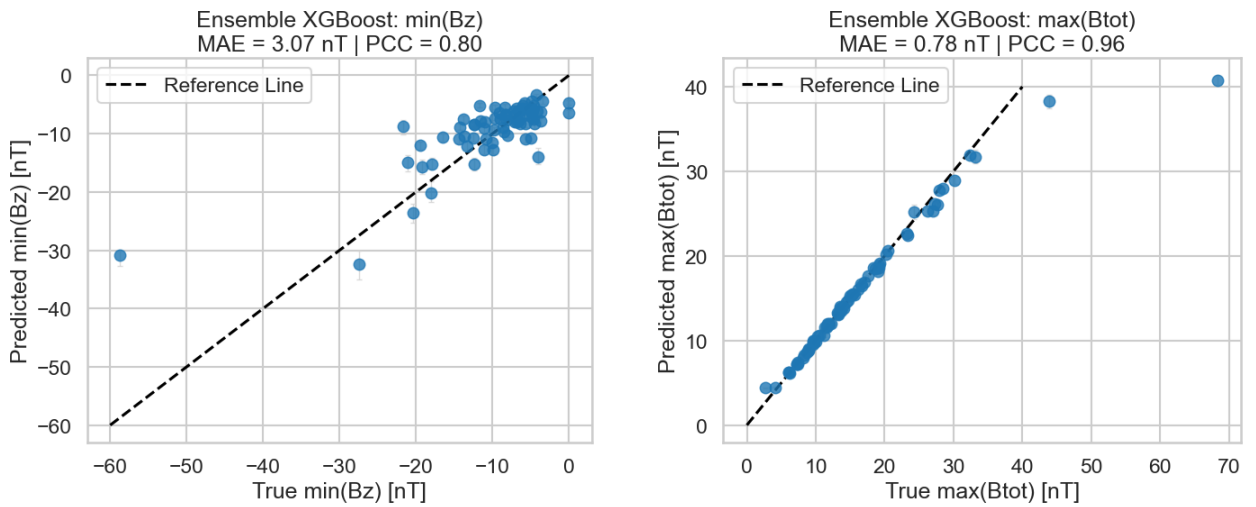


Figure 4.4: Ensemble predictions with 10 - 90% confidence bands. Each point represents the mean prediction across 20 models, with vertical lines indicating uncertainty intervals. The high correlation and narrow bands confirm the ensemble's stability.

Chapter 5

Comparison and Conclusions

In this final chapter we will summarize the predictive performance of the three models explored in this thesis: CNN, XGBoost, and ensemble XGBoost. In addition, we will compare them with the benchmark results reported by [11]. Each model was evaluated on a held-out test set using the same input features and stratified sampling strategy, allowing for a fair comparison across techniques.

Model Comparison

Model	Target	MAE (nT)	RMSE (nT)	PCC	Skill Score
Reiss et al. (2021)	$\min(B_z)$	3.12	4.77	0.71	0.49
	$\max(B_{\text{tot}})$	2.23	3.20	0.91	0.82
CNN (2-layer)	$\min(B_z)$	3.88	5.24	0.76	0.55
	$\max(B_{\text{tot}})$	1.10	1.98	0.98	0.96
XGBoost (GBR)	$\min(B_z)$	3.08	4.82	0.80	0.36
	$\max(B_{\text{tot}})$	0.79	3.29	0.96	0.88
Ensemble XGBoost	$\min(B_z)$	3.07	4.84	0.80	0.36
	$\max(B_{\text{tot}})$	0.78	3.34	0.96	0.88

Table 5.1: *Performance comparison between models developed in this work and those reported by [11]. PCC = Pearson correlation coefficient.*

The results show that all three models were able to replicate or surpass the benchmark scores from [11], despite using the same 42 handcrafted statistical features. The CNN performed especially well on predicting the maximum total magnetic field, achieving a PCC of 0.98, well above the 0.91 reported in the original study. This is likely due to the smoother distribution of $\max(B_{\text{tot}})$, which is easier to learn from.

For $\min(B_z)$, the best performance came from the tree-based models, which outperformed both the original benchmark and the CNN in terms of MAE and PCC. Although the Skill Score was slightly lower than in [11], the improved accuracy and stronger linear relationship suggest that boosted decision trees are particularly well suited for modeling asymmetric or noisy targets like B_z , especially when combined with stratified sampling to balance the training distribution.

The ensemble of XGBoost regressors introduced uncertainty estimation without compromising accuracy. While the average prediction performance remained nearly identical to the single GBR, training multiple models allowed us to capture the variability in their outputs. This made it possible to construct empirical confidence intervals for each prediction, offering not just a point estimate, but a realistic range of plausible outcomes, which is especially valuable for operational forecasting.

Conclusion

This thesis replicated and extended the machine learning pipeline proposed by Reiss et al. (2021) for forecasting the geoeffective magnetic field components within ICMEs. By introducing stratified sampling and experimenting with both deep learning and ensemble techniques, we achieved competitive results with improved interpretability and robustness.

While CNNs showed strong performance on smoother targets, tree-based models remained the most effective for the highly variable B_z component. The addition of ensembling provided a path toward probabilistic forecasting, an important next step for space-weather operations.

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